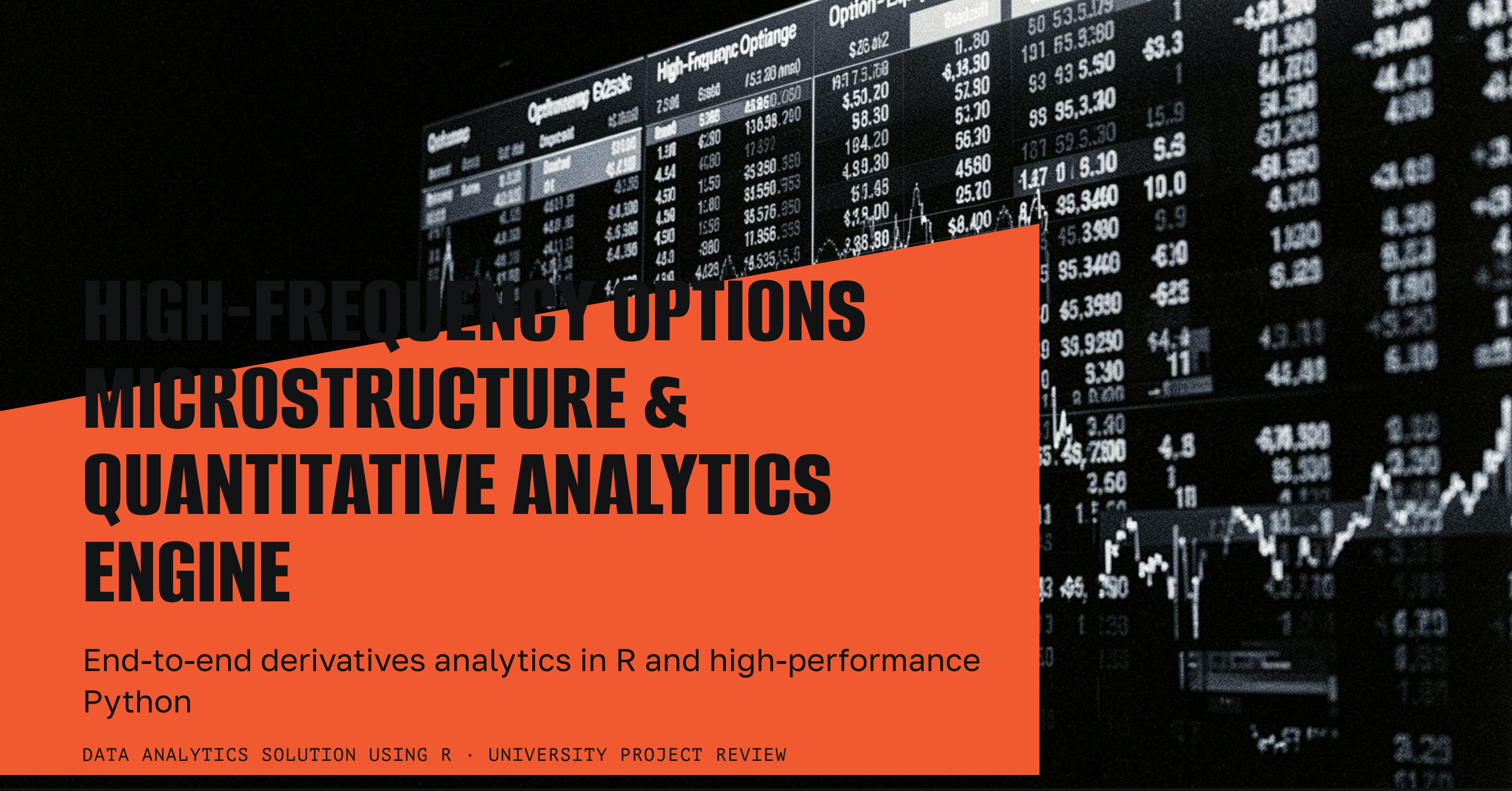


HIGH-FREQUENCY OPTIONS MICROSTRUCTURE & QUANTITATIVE ANALYTICS ENGINE

End-to-end derivatives analytics in R and high-performance Python

DATA ANALYTICS SOLUTION USING R · UNIVERSITY PROJECT REVIEW

RECONSTRUCTION · options microstructure, dealer hedging signals, and quantitative execution analytics

Made with GAMMA

THE PROJECT REPLACES LAGGING INDICATORS WITH DEALER-HEDGING SIGNALS



Backward-looking inputs

Retail indicators such as RSI, moving averages, and MACD look backward and can mislead traders in range-bound markets.



Structural pressure

Track market-maker hedging pressure through Delta, Gamma, and Vanna to detect structural imbalances earlier.

Primary objective: build an institutional-grade, end-to-end quantitative execution platform in R.

Made with GAMMA

THE PROJECT IS EVALUATED ON BOTH ANALYTICS AND ENGINEERING RIGOR

60 MARKS

Evaluation total

Problem definition, data quality, EDA, statistics, modeling, code quality, innovation, documentation, presentation, and demo.

5 TOOLS

Reproducible workflow

tidyverse, dplyr, ggplot2, caret, and rmarkdown with strict GitHub practices.

FACULTY GUIDE

Dr. Vijayalakshmi A, Associate Professor, VIT Chennai.

Made with GAMMA

TWO LIVE FEEDS CAPTURE MARKET STATE AND ORDER-BOOK PRESSURE



Upstox WebSocket V2

Streams Level 2 top-of-book ticks through a persistent Protobuf binary feed.





Direct NSE India HTTP

Polling retrieves option-chain data asynchronously.

62

active instruments across NIFTY, BANKNIFTY, and underlying indices

3 SEC

tick-stream flush interval to Apache Parquet

<340 MS

monitored end-to-end latency

OAuth 2.0 and TOTP refresh authentication daily.

Made with GAMMA

THE DATASET COMBINES 45 SESSIONS AND ABOUT 69.8 MILLION TICKS

45

active market sessions

June 18 to October 11, 2026

69.8M

ticks across the captured sessions

About 69.8 million observations

56,966

Parquet files organized by session date

Core fields

Timestamps, underlying, expiry, strike, option type, spot, LTP, bid, ask, volume, open interest, and latency.

Timing integrity

Exchange timestamps are retained alongside ingestion timestamps to measure network delay.

Made with GAMMA

SEVEN PREPROCESSING STEPS TURN IRREGULAR TICKS INTO MODEL-READY FEATURES

01

Deduplicate ticks

Use a composite key and retain the latest record.

02

Join cash-index spot

Join spot backward in time onto option contracts.

03

Calculate flow

Create incremental volume and open-interest flow.

04

Resample bars

Build 1-minute, 5-minute, and 15-minute bars.

05

Transform strikes

Convert strikes into moneyness and ATM distance.

06

Normalize intraday flow

Use median/IQR time-of-day Z-scores clipped to [-10, +10].

07

Enforce the schema

Validate the feature schema before execution and modeling.

Made with GAMMA

EDA MAKES LIQUIDITY, POSITIONING, AND TELEMETRY VISIBLE

Positioning

Open-interest distributions across strikes compare Calls and Puts and identify the reported Max Pain zone near 24,350.

Liquidity

Bid-ask spreads tighten near ATM and widen sharply in far OTM options, supporting the automated spread penalty filter.

Telemetry

Median transit latency is near 340 ms, with 95% of packets below 490 ms.

Use native charts only for values supported by the project data; do not invent additional observations.

Made with GAMMA

BLACK-SCHOLES, GAMMA, AND VANNA CONNECT PRICE TO HEDGING BEHAVIOR



Black-Scholes-Merton

Estimates structural fair value and exposes gaps between model price and exchange LTP.



Gamma

Measures the rate of Delta acceleration and identifies high-convexity contracts and 0DTE risk.



Vanna

Captures the cross-effect of price and volatility and explains dealer re-hedging during volatility spikes.

Execution layer: Numba LLVM compilation accelerates the pricing engine for high-throughput execution analytics.

Made with GAMMA

REGIME DETECTION SWITCHES THE STRATEGY BEFORE CONDITIONS CHANGE

Trailing 5-minute and 15-minute price velocity plus implied-volatility skew produce a regime-confidence score from 0 to 100.



Momentum breakout mode

Activates above 60 confidence and targets high-Gamma directional buying with trailing stops.

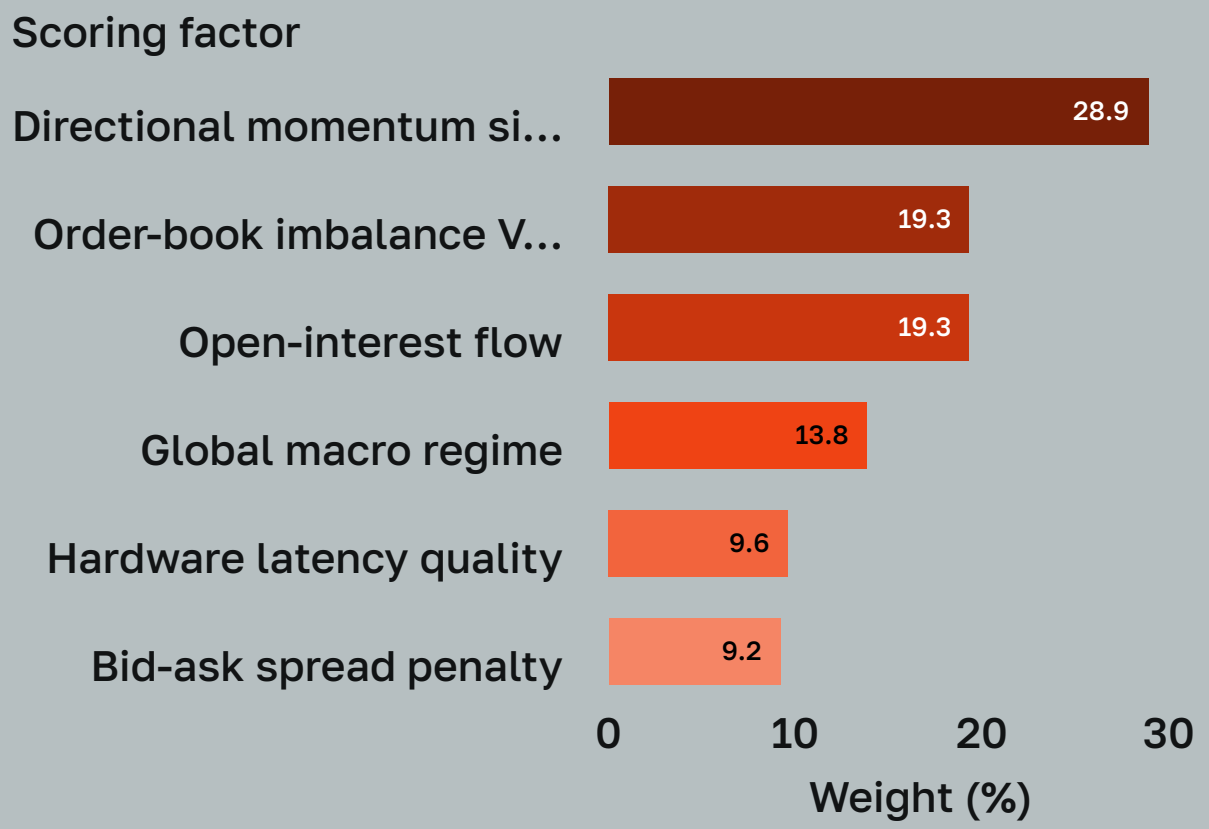
Mean-reversion mode

Bans directional buying in choppy markets and shifts to defined-risk credit spreads to harvest premium decay.

Made with GAMMA

09 · EXECUTION FILTER

THE SCORING MATRIX BLOCKS WEAK AND EXPENSIVE TRADES



IGNAL MEETS FRICTION.

The combined score rejects stale, illiquid, low-conviction, and negative-EV candidates.

Made with GAMMA